

COMP90086 Group Project

Jason Kang
Master of IT (Computing)
The University of Melbourne
995465

Abstract— This research explores the utilization of Siamese Networks (that uses discriminative features to learn embeddings for each input, which are then compared to determine similarity or dissimilarity between samples) with the triplet loss function to predict the top-2 most similar images from a selection of candidate images in the "Totally Looks Like" dataset. The study evaluates the performance of two pre-trained models, ResNet50 and VGG16, and combines generic and face features to determine image similarity. Our most accurate model, merging ResNet50 and VGG16, demonstrates significant accuracy in split testing and Kaggle public scoring. This research highlights the efficacy of Siamese Networks and pre-trained models in improving image similarity predictions for computer vision applications.

Keywords: *Computer Vision, Totally Looks Like, Machine Learning, Face Recognition, Siamese Network, VGG16, ResNet50, Triplet Loss, MTCNN*

I. INTRODUCTION

As the information technology age develops, there is an increased amount of activity online. As the traffic online develops, the disparity between humans and AI becomes increasingly reduced especially in the context of computer vision. This report has utilised the task of the Totally Looks Like challenge [1] to emphasise the difference, or lack of, in performance between human and AI. To achieve this, we employ Siamese Networks, pre-trained Convolutional Neural Networks (CNNs), and the triplet loss function. We use these tools to investigate AI's ability to distinguish image similarities and uncover the challenges and nuances that emerge in the process.

Our experiment involves the utilization of prominent CNN models, ResNet50 and VGG16, to extract meaningful image features. These models serve as the foundation of our exploration, enabling us to assess their performance in the challenging task of image comparison.

Intriguingly, we examine how the combination of both generic and face features, extracted from the candidates, can enhance the accuracy of similarity predictions. The journey through the nuances of computer vision and machine learning leads to a nuanced understanding of how AI fares in discerning the subtle differences and similarities between images.

This report chronicles our findings and insights, shedding light on the interactions between technology and human perception, while striving to achieve the goal of replicating the remarkable capabilities of the human eye in the digital realm.

II. METHODOLOGY

A. Dataset

The dataset was obtained from the "Totally Looks Like" challenge [1]. It includes a substantial number of images, enhancing the comprehensiveness of our research.

For this research 2000 images were divided into a training set and a validation set for model optimization before testing on a separate test set. 1800 images were used for training and 15% were randomly chosen to be used as validation for the model. Then these were tested on 200 images. Additionally, the negative images in the dataset were replicated tenfold to create additional negative images and train more to minimise the triplet loss.

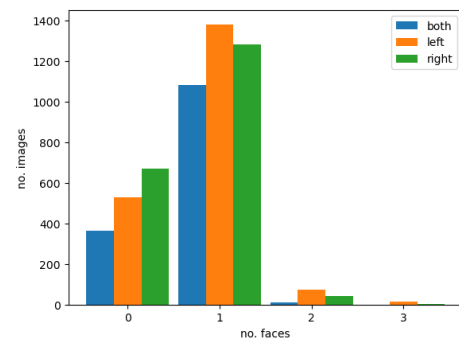


Figure 1: Number of faces in dataset

B. Siamese Network

A Siamese network [2] is a neural network that finds the similarity between 2 pairs of input. It consists of two identical subnetworks that have identical architecture and parameters, and these subnetworks process the inputs separately. However, the weights and parameters of said networks are shared and ensure that both networks extract and learn the same feature representation. Once the feature representation is modelled, the distance between the vectors can be calculated and the model is trained to minimise the loss function. When training, the triplet loss function was used.

C. Pre Trained Convolutional Networks

Pre trained Convolutional Neural Networks (CNN) were leveraged to be used as the subnetworks in the Siamese network, specifically Visual Geometry Group 16) and Residual Network 50 (ResNet50) from Keras to extract feature representations from the dataset. These models were trained on ImageNet by separate entities. Furthermore, on the final layers of the CNN a dense layer was added to learn and enhance the feature embedding.

a) Visual Geometry Group 16 (VGG16)

Developed at the University of Oxford, this model [3] has a total of 16 weight layers and employs 3x3 filters which is then followed up by ReLU activation and max pooling. To find the most similar image between the lookalikes, VGG16's feature extraction was harnessed.

b) Residual Network 50 (ResNet50)

Residual Network 50 [4] was introduced by Microsoft Research and uses a residual connection to have a deep network. The network has 50 weight layers organised into 16 residual blocks that are composed of convolution, batch normalisation, ReLU activation and skip connections.

D. Triplet loss function

The triplet loss function operates by having a triplet, an anchor, positive data point and a negative data point. The anchor is a reference data point that the model wants to learn, the positive point is the point (most) similar to the anchor whereas the negative is (most) dissimilar.

The goal is to minimise the distance between the anchor and positive data points while maximising the distance between the anchor and the negative data points.

$$L(a, p, n) = \max(\|f(a) - f(p)\|^2 - \|f(a) - f(n)\|^2 + \text{margin}, 0) \quad (1)$$

Where $L(a, p, n)$ is the loss for the triplet and $f(a)$, $f(p)$ and $f(n)$ are the feature vectors.

$\|f(x)\|^2$ is the squared Euclidean distance and the margin defines the absolute minimum distance between the anchor to positive and anchor to negative.

E. Euclidean distance and Sum of Squares

The sum of squares refers to the squared Euclidean distance between a pair of feature vectors. A triplet is first selected, and these are passed through the Siamese network to obtain the feature embeddings which are then represented in the feature dimension.

Then the squared Euclidean distance is calculated between the anchor and positive and the anchor and negative by the following formulas

$$d_{\text{pos}}^2 = \|f(\text{anchor}) - f(\text{positive})\|^2 \quad (2)$$

$$d_{\text{neg}}^2 = \|f(\text{anchor}) - f(\text{negative})\|^2 \quad (3)$$

Where $f(x)$ represents the feature embedding of the image.

Then this is used to minimise the difference between those two in the triplet loss function and this guides the network to learn the dimensional space of features and what makes similar pairs close and dissimilar pairs far apart.

F. Cosine Distance

Often referred to as cosine similarity where in this project, features were extracted using pre-trained CNN models. Then the images were represented as a vector with each dimension corresponding to a feature. When we compare images cosine similarity was used after the model was trained.

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} \quad (4)$$

Where:

A and B are the vectors, you want to compare. $A \cdot B$ is the dot product of the vectors A and B.

$\|A\|$ and $\|B\|$ represent the Euclidean norms (magnitudes) of the vectors A and B, respectively.

The cosine distance is then calculated as $1 - \text{cosine similarity}$ to convert the similarity to a distance metric.

G. MTCNN

MTCNN (Multi-Task Cascaded Convolutional Neural Networks) is a deep learning framework which detects faces and facial landmarks on images proposed by [5]. It consists of three stages: face detection, bounding box refinement, and facial landmark localization. Using a cascaded approach, MTCNN progressively refines its predictions, making it robust in detecting faces under various conditions. It has found wide applications in face-related tasks due to its accuracy and adaptability, making it a fundamental component in computer vision applications.

III. EXPERIMENTAL OUTCOMES & SIGNIFICANCE

The proposed model was developed using the Siamese Network to generate embeddings for each of the images of the triplet. Then those embeddings were separated using ResNet50 by adding 3 custom dense layers. The flatten layer is employed to flatten the output from the base model before passing it through the custom dense layers. The dense layers 'Dense(512, activation="relu")' and 'Dense(256, activation="relu")' are responsible for learning complex patterns and relationships in the extracted features. To prevent overfitting and ensure faster convergence during the training process, batch normalization was added after each dense layer making the model more stable and allowing for higher learning rates. To avoid affecting the weights of the pretrained model, the layers until conv5_block1_out layer was frozen, and the bottom few layers were left dynamic for fine tuning purposes.

Model	Accuracy (Cosine distance)
ResNet50 Generic	0.5628

a. Accuracy of ResNet50 Model on test set

With an accuracy of 0.5628 ResNet50 was able to predict successfully to some extent and tended to match anchor images with the candidate image that contained the person's face because most of the training set consists of an anchor and positive face pairs.

To understand whether ResNet50 was underperforming and determine whether the model's parameter count, and depth correlates with accuracy VGG16 was utilised. The same custom layer was used but the weights of all layers in both models were frozen until block5_conv3 layer and again left the bottom few layers trainable for fine-tuning purposes. VGG16 was chosen for its simpler architecture making it faster to train and as the architecture differs along with its training technique it was expected to improve performance as seen in the table below.

Model	Parameters	Depth	Top-1 Accuracy	Top-5 Accuracy
ResNet50	25.6M	107	71.3%	90.1%

VGG16	138.4M	16	74.9%	92.1%
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b. Difference in architecture and performance between the two models

Model	Accuracy
Resnet50 Generic	0.5628
VGG16 Generic	0.5377

c. Accuracy of Generic Models on test set

VGG16 underperformed contrary to expectations, and this may be a result of the shallower architecture.

When considering the size of the dataset and its smaller nature, larger number of parameters in VGG16 makes it prone to overfitting. In contrast, architecture like ResNet50 is more parameter efficient, leading to better generalisation and performance.

So, for the next model 10 negative images were used for training rather than 1. A more ideal approach would be to select the most similar image to the ground truth and using it as the negative image but due to a lack of computational resources it was and still is infeasible. The run time for 1800 pairs of training images it would have to compute the similarity of the other 1798 images which is 1798×1800 iterations.

Accuracy with 1 negative image	Accuracy with 10 negative images
0.5628	0.6131

d. Accuracy of ResNet50 Generic Model on test set

The table above outlines the improvements when using 10 negative images and this is because of greater generalisation as the increased negative samples exposes the model to a wider range of dissimilar pairs. Thus, making it more robust to variations in the dataset, reducing overfitting and balancing the dataset.

In consideration of the dataset and the primary objective of identifying 'lookalikes,' the focus of this report shifted to faces. To extract relevant face regions from the left and right pair images, the MTCNN was employed. However, given the relative complexity of the MTCNN model, we had to carefully balance accuracy and computational cost.

To address these challenges, the size of the extracted face regions was modified to $128 \times 128 \times 3$, reducing the computational demand while making some trade-offs in terms of accuracy. Subsequently, these modified face regions were utilized for training both VGG16 and ResNet50 models. The training parameters remained consistent with the previous setup, ensuring a fair comparison between the two architectures."

Model	Accuracy
VGG16 Face	0.5047
Resnet50 Face	0.4859

e. Accuracy of VGG16 and ResNet50 Face Models on test set

This resulted in a lower accuracy because of using a model specifically designed for identifying just faces in the dataset and as VGG16 and ResNet50 are already pretrained it won't have the dimensional space required for the totally look alike dataset.

We then considered the dataset as a whole and visually identified where predictions were made incorrectly, and it was noticed that the generic model often predicted the face of the same person incorrectly.

		
Anchor	False Negative	False Positive

f. misclassified image by Face Models

Where the two males have similar hair style, nose shape, chin and even a collar on both outfits. However, it predicted the wrong image possibly due to the direction the males are facing or the location of the light source in the images.

So, the VGG16 face model was combined with the ResNet50 generic model to rank the images in similarity. The evaluation metric, top 2 accuracy, from the project was used and treated the most similar image of each model as a top-2 accuracy.

Model	Accuracy
Hybrid	0.7387

g. Accuracy of VGG16 Face Model combined with ResNet50 Generic Model on test set

The result was an improvement in accuracy and allows for better data representation as MTCNN is designed for facial detection and alignment, so those features were standardised. So, the Siamese network received higher quality input data which improved the model's ability to distinguish the features between the faces. Additionally, noisy regions from the image are removing unnecessary information for the Siamese network to process so finetuning the network with MTCNN processed data was successful to some extent.

The models presented in this experiment were tested in a competition on Kaggle [6] but got a lower accuracy. This could be due to several different reasons, from data distribution being different, our model has overfit to the training data, our parameter tuning isn't optimal for the Kaggle test set or the Kaggle test set is more challenging compared to the original test set.

Model	Kaggle Accuracy
VGG16 Generic	0.4
Resnet50 Generic	0.432
Face (VGG16) and Generic (Resnet50)	0.441

IV. RECOMMENDATIONS

Some learnings from this experiment were that firstly the computation time must be considered and exploring techniques to optimise the model's run time would allow for more results and experimentation. Moving on from this, feature engineering is key to this experiment and although faces where the main feature other features could've been focused on as the lookalike challenge is not limited to faces and combining these features with deep learning will enhance the model's ability to identify image similarities. Additionally, the models can be explored and adjusted as their architecture differs and the learning rate, dropout rates and batch size can be tuned to find the optimal configuration for this task. Different layers could also be used in the architecture and finding the best configuration will lead to better results. The loss function in the Siamese network can also be optimised by changing the margin and finding the margin that best matches with the dataset. Extending beyond the networks and models, more models could've been used to boost accuracy and robustness.

However, the greatest improvement can be made by doing a thorough analysis on misclassified images. As observed in the table below, the facial expression, shape and texture are obvious to the human's eye, but the model cannot recognise these features and engineering the model to identify these features will boost accuracy.

Anchor	False Negative	False Positive
		
		

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i. Accuracy of models on Kaggle test set

V. CONCLUSION

In this project, we delved into the utilization of Siamese Networks with the triplet loss function to predict the top-2 most similar images from a pool of 20 candidate images within the "Totally Looks Like" dataset. To address this challenge, we conducted a comparative analysis of two pre-trained models, namely ResNet50 and VGG16, harnessing both generic and face features to facilitate a collaborative voting mechanism for identifying the top-2 most similar images. Our most accurate model, which combined ResNet50 as the generic model and VGG16 as the face model, demonstrated an impressive accuracy of 73.87% when evaluated on a split testing dataset. Furthermore, when assessed using Kaggle's public score, the model exhibited a noteworthy accuracy of 44.1%. These findings underscore the potential of Siamese Networks and pre-trained models in enhancing image similarity predictions.

VI. REFERENCES

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